Lecture 4: detailed solutions

Real book exercises — Economics of Organizations and Incentives (MPE Insper)

Solutions to `Listas/lista-reais-lecture-04.pdf`

November 2026

These are detailed solutions to the real book exercises of the list `lista-reais-lectures-01-05.pdf` in `Listas/` (one block of ten per lecture). Within each lecture the items run from easy to hard, in the list's order, and each heading shows the item's number in the lecture, its number in the combined list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Statements are reproduced verbatim in the list and are not repeated here. Every solution has the same six parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide; (c) the solution, every step shown (for matrices, the full dominance comparisons or best-response marking); (d) the economic interpretation; (e) the verification, by the Python check scripts in this folder and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG) and Cabral (2017). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of IESDS with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $\sigma_i \in \Delta(S_i)$ is a mixed strategy. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

Contents

Lecture 4 — Credible threats and subgame perfection **3**

Item 1 (31): Gibbons Problem 2.12 (p. 135) **3**

Item 2 (32): Tadelis Exercise 7.6 (p. 148–149) **5**

Item 3 (33): Tadelis Exercise 8.8 (p. 171–172) **7**

Item 4 (34): Tadelis Exercise 8.6 (p. 170–171) **9**

Item 5 (35): Muñoz-Garcia Ch. 4, Exercise 11 (p. 180) **11**

Item 6 (36): Tadelis Exercise 8.11 (p. 172–173) **14**

Item 7 (37): Tadelis Exercise 8.4 (p. 170) **17**

Item 8 (38): Tadelis Exercise 8.13 (p. 173–174) **19**

Item 9 (39): Tadelis Exercise 8.7 (p. 171) **22**

Item 10 (40): MWG Exercise 9.B.5 (p. 302) **26**

Lecture 4 — Credible threats and subgame perfection

Item 1 (item 31 of the combined list)**: Gibbons Problem 2.12** Foundations · difficulty 1/5

Source: Gibbons Problem 2.12, page 135. The statement is reproduced verbatim in the list (item 31); it is not repeated here.

(a) What is asked. Define a strategy, a subgame and a subgame-perfect Nash equilibrium in a repeated game.

(b) Method. The definitions of Gibbons §2.3.B (pp. 93–95), stated for the finitely repeated game $G(T)$ and the infinitely repeated game $G(\infty, \delta)$; the matching definitions in the slides (Lecture 4, slides 21–22: repeated game, $G(T, \delta)$, "a strategy says what to play in each period after each possible history", "each period starts a new subgame"; SPE, slide 17) and in Tadelis §9.3–9.4 (pp. 178–182). Each definition is illustrated with the two-stage Prisoners' Dilemma of Gibbons Fig. 2.3.1 (p. 82).

(c) Solution. Setting. A stage game $G = \{A_1, \dots, A_n;\ u_1, \dots, u_n\}$ is played in periods $t = 1, 2, \dots$; before each period all players observe all past actions. The *history* through period t is $h^t = (a^1, \dots, a^t)$, with $a^s = (a_1^s, \dots, a_n^s)$ the action profile of period s; $h^0 = \varnothing$. H^t is the set of histories through t.

1. Strategy. A strategy of player i specifies the action he will take in each stage, for each possible history of play through the previous stage (Gibbons p. 94):

$$s_i = (s_i^1, s_i^2, \dots), \qquad s_i^t : H^{t-1} \to A_i.$$

It is a complete contingent plan, including histories that the plan itself would never produce. *Example.* In the two-stage Prisoners' Dilemma (actions L_i, R_i) there are four first-stage outcomes, so a strategy has five instructions (v, w, x, y, z): a first-stage action and a second-stage action after each of (L_1, L_2), (L_1, R_2), (R_1, L_2), (R_1, R_2) (Gibbons p. 93). Each player has $2^5 = 32$ pure strategies. In general, with a 2×2 stage game played T times, player i has $1 + 4 + \cdots + 4^{T-1} = \frac{4^T - 1}{3}$ information sets and $2^{(4^T-1)/3}$ pure strategies.

2. Subgame. A subgame is the piece of the game that remains to be played from any point at which the complete history so far is common knowledge (Gibbons p. 94). In $G(T)$ a subgame beginning at stage $t+1$ is the repeated game in which G is played $T - t$ times, $G(T - t)$, following one particular history h^t; there is one such subgame for each $h^t \in H^t$. In $G(\infty, \delta)$ each subgame beginning at stage $t + 1$ is identical to $G(\infty, \delta)$ itself, and again there is one for each history through t. *Not* a subgame: stage t taken on its own, for $t < T$; a subgame includes all the moves that follow its starting point (Gibbons p. 95). *Example.* The two-stage Prisoners' Dilemma has four proper subgames after period 1 (one per first-stage outcome), each a one-shot Prisoners' Dilemma, plus the whole game.

3. Subgame-perfect Nash equilibrium. A Nash equilibrium is subgame-perfect if the players' strategies constitute a Nash equilibrium in every subgame (Selten 1965; Gibbons p. 95; Lecture 4, slide 17 states it for general extensive forms: the restriction of the profile to every proper subgame is a Nash equilibrium of that subgame). In a repeated game this means: after *every* history h^t, on or off the equilibrium path, the continuation strategies $s|_{h^t}$ form a Nash equilibrium of the continuation game.

Example (computed). In the two-stage Prisoners' Dilemma ($u(L,L) = (1,1)$, $u(L,R) = (5,0)$, $u(R,L) = (0,5)$, $u(R,R) = (4,4)$, payoffs summed over the two stages), enumerating all 32×32 profiles gives 16 Nash equilibria; all of them have the outcome (L_1, L_2) in both stages, payoffs $(2,2)$, but they differ in what is planned after unreached first-stage outcomes. Exactly one is subgame perfect: each player plays L in stage 1 and L after every history, (L, L, L, L, L), because each of the four second-stage subgames is a one-shot Prisoners' Dilemma whose only Nash equilibrium is (L_1, L_2).

(d) Interpretation. Three ideas carry over from Lectures 3–4 to repeated games. A strategy must say what happens after deviations, because those plans are what make promises and threats; a subgame starts after each observed history; and subgame perfection requires the punishments and rewards to be ones players would actually carry out. This is the toolkit for the carrots-and-sticks and folk-theorem arguments of Lecture 5: cooperation can be sustained only by continuation play that is itself an equilibrium.

(e) Verification. `check_B_L04.py` enumerates the 32 strategies (v, w, x, y, z) of each player in the two-stage Prisoners' Dilemma, computes all Nash equilibria of the 32×32 game and keeps those that play a stage Nash equilibrium in every second-stage subgame; it also checks the count $\frac{4^T-1}{3}$ for $T = 1, 2, 3$ and the 3^{10} strategies of the 3×3 two-stage game of Gibbons Fig. 2.3.3. Output line: `gibbons-2-12: 32 strategies each, 16 NE profiles, 1 SPE (L,L,L,L,L) each`. Gibbons prints no separate answer; the definitions are those of pp. 94–95.

(f) Common mistakes.

- Defining a strategy as a sequence of actions, one per period ("play b, then c"): it must be contingent on the history.
- Calling the stage game played in period t a subgame: a subgame includes all later periods.
- Checking Nash behavior only on the equilibrium path: subgame perfection is about every history, including those after deviations.

Answer. A strategy assigns an action in every period to every possible history of play through the previous period. A subgame is the remainder of the repeated game after a given history (in $G(T)$, the game $G(T-t)$ after one history h^t; in $G(\infty, \delta)$, a copy of the whole game). A subgame-perfect Nash equilibrium is a strategy profile whose restriction to every subgame, after every history, is a Nash equilibrium of that subgame.

Item 2 (item 32 of the combined list)**: Tadelis Exercise 7.6** Foundations · difficulty 1/5

Source: Tadelis Exercise 7.6, page 148–149. The statement is reproduced verbatim in the list (item 32); it is not repeated here.

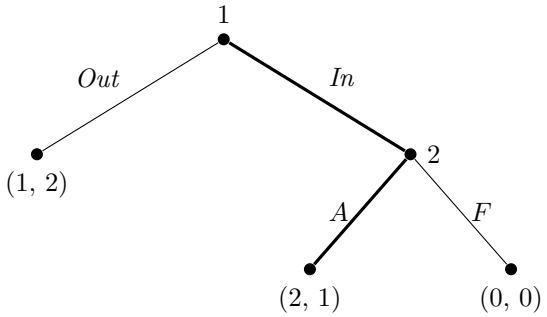
(a) What is asked. (a) A game tree with payoffs that represent the stated rankings; (b) the number of pure strategies of each player; (c) all Nash equilibria.

(b) Method. Extensive form and game trees (Tadelis §7.1, pp. 130–137); pure strategies as complete plans (§7.2.1, p. 137); the normal-form representation of a game tree (§7.2.3, p. 143); Nash equilibrium and paths of play, on and off the equilibrium path (§7.3 and Def. 7.8, pp. 145–146). The question stops at Nash equilibrium; which equilibrium survives sequential rationality is Tadelis §8.1 (Def. 8.1, p. 152) and Lecture 4, slides 13 and 17 (the entry-deterrence logic of the Brexit case, slides 9–11).

(c) Solution. (a) Tree. Ordinal payoffs suffice, since only the rankings are given. Write *In* and *Out* for player 1's choice and A (accommodate), F (fight) for player 2's. The rankings are

$$\text{player 1: } (In, A) \succ Out \succ (In, F), \qquad \text{player 2: } Out \succ (In, A) \succ (In, F).$$

Player 1's middle outcome must be *Out* and player 2's must be (In, A), the only outcomes not named as best or worst. Payoffs $2, 1, 0$ in that order give



Payoffs (u_1, u_2); thick branches = backward-induction choices (used below).

Any numbers with the same rankings give the same answers to (b) and (c).

(b) Pure strategies. Each player has one information set with two actions, so $S_1 = \{In, Out\}$ and $S_2 = \{A, F\}$: two pure strategies each. Player 2's strategy is a plan for the node after entry, and it is part of the profile even when player 1 stays out.

(c) Nash equilibria. The normal form, with best replies underlined:

	A	F
In	$\underline{2}, \underline{1}$	$0, 0$
Out	$1, \underline{2}$	$\underline{1}, \underline{2}$

Pure. (In, A) and (Out, F). Against A, *In* gives $2 > 1$; against F, *Out* gives $1 > 0$. Against *In*, A gives $1 > 0$; against *Out*, player 2 gets 2 with either action, so F is a (weak) best reply.

Mixed. Let $p = \Pr(In)$ and $f = \Pr(F)$.

- If $p > 0$, then $u_2(A) - u_2(F) = p \cdot (1 - 0) > 0$, so $f = 0$, and then *In* is player 1's unique best reply $(2 > 1)$: $p = 1$. This is (In, A).

- If $p = 0$, player 2 is indifferent, so any f is a best reply. *Out* is a best reply for player 1 iff

$$u_1(Out) = 1 \;\geq\; u_1(In) = 2(1-f) + 0 \cdot f \iff f \geq \tfrac{1}{2}.$$

So the set of all Nash equilibria is

$$\{(In, A)\} \cup \{(Out,\ \sigma_2) : \sigma_2(F) \geq \tfrac{1}{2}\},$$

a single entry equilibrium plus a continuum of deterrence equilibria, (Out, F) among them.

Which are sequentially rational. At player 2's node, A gives $1 > 0$. Every equilibrium of the second family puts weight $f \geq \frac{1}{2}$ on F at a node that is off the equilibrium path (Def. 7.8), where F is not a best response. Backward induction (thick branches) keeps only (In, A), which is therefore the unique subgame-perfect equilibrium (Tadelis Fact, p. 158; slide 17).

(d) Interpretation. "Fight if you enter" deters entry only as long as nobody tests it. Once the entrant is in, the price war hurts the incumbent ($0 < 1$), so the threat is not credible, and an entrant who reasons backward enters. Nash equilibrium cannot see this: in (Out, F) the war is never fought, so planning it costs player 2 nothing. To deter entry, the incumbent needs a commitment made before the entry decision that makes fighting its best response after entry (excess capacity, a sunk investment, a contract). The Brexit case uses the same logic: the UK's sunk preparation makes rejecting a bad deal credible (slides 9–11).

(e) Verification. `check_D.py` builds the tree, enumerates the 2×2 pure profiles, tests every one for Nash and for subgame perfection by definition, and tests every mixed profile on a $\frac{1}{100}$ grid in both probabilities with exact fractions. Output: "Tadelis 7.6: 2 strategies each; pure NE (In,A), (Out,F); all NE = (In,A) and Out with Pr(F) >= 1/2; SPE (In,A) only". Tadelis prints no answer.

(f) Common mistakes.

- Giving player 1 the payoff ordering of a "most-preferred / least-preferred" list without placing *Out* in the middle. The middle outcome is implied, not stated.
- Answering (c) with (In, A) only. The question asks for Nash equilibria, and (Out, F) is one, as are all (Out, σ_2) with $\sigma_2(F) \geq \frac{1}{2}$.
- Calling (Out, F) not Nash because "player 2 would not fight". Player 2 is indifferent on the path; the flaw is off the path, which is what subgame perfection adds.

Answer. (a) Tree above, with $u(Out) = (1, 2)$, $u(In, A) = (2, 1)$, $u(In, F) = (0, 0)$. (b) Two pure strategies each: $\{In, Out\}$ and $\{A, F\}$. (c) Pure Nash equilibria (In, A) and (Out, F); with mixing, all Nash equilibria are (In, A) and (Out, σ_2) with $\sigma_2(F) \geq \frac{1}{2}$. Only (In, A) is subgame perfect: the threat to fight is not credible.

Item 3 (item 33 of the combined list)**: Tadelis Exercise 8.8** Foundations · difficulty 2/5

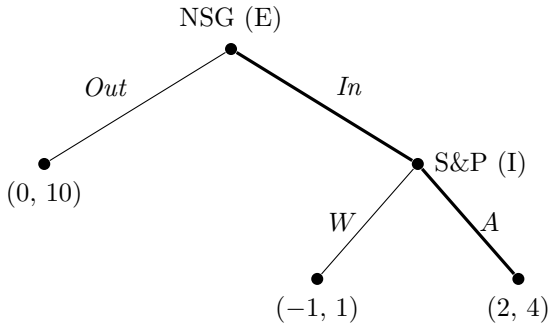
Source: Tadelis Exercise 8.8, page 171–172. The statement is reproduced verbatim in the list (item 33); it is not repeated here.

(a) What is asked. Set up the game tree of the entry story (NSG enters or not, then S&P fights or accommodates) and find the subgame-perfect equilibrium outcome.

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slide 30; Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 152–153). With perfect information the subgame-perfect equilibria are exactly the profiles that survive backward induction (Lecture 4, slide 17; Tadelis §8.2, p. 157). Non-credible threats in Nash equilibrium: Lecture 4, slides 8 and 13. Same structure as the entry game of Cabral §7.2 (`Notas/L03/L03-06`).

(c) Solution. Notation. Players: NSG (the entrant, E) and S&P (the incumbent, I). E chooses *In* or *Out*; after *In*, I chooses W (price war) or A (accommodate). Payoffs (u_E, u_I) in \$ million of net present value. The entry cost of 2 is sunk once E enters, so it is subtracted from every payoff that follows *In*:

$$\begin{aligned}(u_E, u_I)(Out) &= (0,\ 10),\\ (u_E, u_I)(In, W) &= (1-2,\ 1) = (-1,\ 1),\\ (u_E, u_I)(In, A) &= (4-2,\ 4) = (2,\ 4).\end{aligned}$$



Payoffs (u_E, u_I); thick branches = backward-induction choices.

Step 1: the incumbent's node. After entry, S&P compares

$$u_I(In, A) = 4 > 1 = u_I(In, W),$$

so it accommodates. The price war is not a credible response.

Step 2: the entrant's node. Replacing S&P's node by its outcome $(2, 4)$, NSG compares

$$u_E(In) = 2 > 0 = u_E(Out),$$

so it enters. SPE: (In, A), outcome entry and accommodation, payoffs $(2, 4)$. The game has two proper subgames (the whole game and the one at S&P's node), and the SPE plays a Nash equilibrium in both.

Step 3: all Nash equilibria. Normal form (rows: E's strategies; columns: I's strategies):

	W	A
Out	$\underline{0},\ \underline{10}$	$0,\ \underline{10}$
In	$-1,\ 1$	$\underline{2},\ \underline{4}$

Pure Nash equilibria: (In, A) and (Out, W). With $w = \Pr_I(W)$, NSG's payoff from entering is $-w + 2(1-w) = 2 - 3w$, so Out is a best reply iff $0 \geq 2 - 3w$, i.e. $w \geq \frac{2}{3}$; S&P is not reached and is indifferent. The full set is

$$\{(In, A)\} \ \cup \ \{(Out, \ w\,W + (1-w)A) : w \geq \tfrac{2}{3}\}.$$

Every equilibrium in the second family fails sequential rationality at the node S&P never reaches (Tadelis Def. 8.1, p. 152): the threat "war with probability at least $\frac{2}{3}$" costs nothing because it is never carried out, and it would not be carried out ($1 < 4$).

Step 4: what would make the threat credible (deepening). Suppose S&P could, before NSG decides, commit to a penalty b that it pays if it accommodates (a public contract, a capacity investment, a "most-favored-customer" clause). Then at its node S&P fights iff $1 > 4 - b$, i.e. $b > 3$; NSG, anticipating war, stays out ($-1 < 0$), and S&P earns 10 without ever paying b. A credible commitment is worth up to $10 - 4 = 6$ to S&P. Independently, an entry cost above 4 would deter entry even with accommodation ($4 - \text{cost} < 0$).

(d) Interpretation. S&P's prediction of a price war is cheap talk: once NSG has sunk its \$2 million, a war gives S&P 1 instead of 4, and NSG knows it. These are the three lessons of the Brexit case (Lecture 4, slide 11) in miniature: *talk is cheap* (announcing war without changing payoffs is not credible); *sunk investments can be strategic* (only a sunk, payoff-changing move such as the penalty b makes war credible); *observability matters* (the commitment works only if NSG sees it before deciding). The outcome does not depend on how the entry cost is booked: if "each firm earns \$1 million" were already net of it, NSG would still enter ($2 > 0$, and even a war would pay $1 > 0$).

(e) Verification. `check_B_L04.py` recomputes both backward-induction steps, the two pure Nash equilibria of the 2×2 normal form, scans w on a grid of step $\frac{1}{300}$ (Nash equilibria exactly at $w = 0$ with entry and $w \geq \frac{2}{3}$ with Out), and checks the commitment thresholds $b > 3$ and the value 6. Output line: `tadelis-8-8: SPE (In, A); NE (Out, Pr(W) >= 2/3); commitment worth up to 6`. Tadelis prints no answer.

(f) Common mistakes.

- Forgetting to subtract the sunk entry cost from NSG's payoffs after entry.
- Reporting (Out, W) as an SPE because it is a Nash equilibrium.
- Giving the outcome only ("NSG enters") when the SPE is a strategy profile: S&P's plan A must be stated.

Answer. Tree above. Subgame-perfect equilibrium: NSG plays In, S&P plays A (accommodate); outcome: entry and accommodation, payoffs $(2, 4)$ in \$ million (NSG net of the entry cost). The other Nash equilibria, Out with $\Pr(W) \geq \frac{2}{3}$, rest on a non-credible threat.

Item 4 (item 34 of the combined list)**: Tadelis Exercise 8.6** Exam · difficulty 3/5

Source: Tadelis Exercise 8.6, page 170–171. The statement is reproduced verbatim in the list (item 34); it is not repeated here.

(a) What is asked. Firm 1 may pay F to cut its unit cost from 10 to 5 before a Cournot duopoly with $p = 100 - q$: (a) the threshold F^* below which the unique SPE has firm 1 investing; (b) for $F > F^*$, a Nash equilibrium that is not subgame perfect.

(b) Method. A two-stage game: an observed investment decision, then a simultaneous Cournot game (Gibbons §2.2.A, pp. 71–73; Tadelis §8.2). SPE: solve the Cournot subgame after each investment decision, then compare at the root (Lecture 4, slide 17; Tadelis Def. 8.3, p. 157). Cournot with asymmetric linear costs: first-order conditions, as in Lecture 2 and Gibbons §1.2.A. Non-SPE Nash equilibria by off-path threats (Lecture 4, slides 8 and 13).

(c) Solution. Notation. Firm 1 chooses I (invest) or N; both firms observe it, then choose $q_1, q_2 \geq 0$. Profits $\pi_i = (100 - q_1 - q_2 - c_i)q_i$, minus F for firm 1 after I.

Step 1: Cournot subgame with costs (c_1, c_2)**.** The first-order conditions are

$$100 - 2q_1 - q_2 - c_1 = 0, \qquad 100 - q_1 - 2q_2 - c_2 = 0,$$

so the best replies are $q_1 = \frac{100 - c_1 - q_2}{2}$, $q_2 = \frac{100 - c_2 - q_1}{2}$, and

$$q_1 = \frac{100 - 2c_1 + c_2}{3}, \qquad q_2 = \frac{100 - 2c_2 + c_1}{3}, \qquad \pi_i = q_i^2. \tag{1}$$

(The last equality uses $p - c_i = q_i$ from the first-order condition.) The subgame equilibrium is unique (the best replies have slope $-\frac{1}{2}$ and cross once).

Subgame	q_1	q_2	p	(π_1, π_2) before F
N: $c_1 = c_2 = 10$	30	30	40	$(900,\ 900)$
I: $c_1 = 5$, $c_2 = 10$	$\frac{100}{3}$	$\frac{85}{3}$	$\frac{115}{3}$	$(\frac{10000}{9},\ \frac{7225}{9})$

Check for I: $q_1 = \frac{100-10+10}{3} = \frac{100}{3}$, $q_2 = \frac{100-20+5}{3} = \frac{85}{3}$, $p = 100 - \frac{185}{3} = \frac{115}{3}$, $p - 5 = \frac{100}{3} = q_1$.

(a) The threshold. Replace each subgame by its value. At the root firm 1 compares $\frac{10000}{9} - F$ (invest) with 900 (not):

$$F^* = \frac{10000}{9} - 900 = \frac{10000 - 8100}{9} = \frac{1900}{9} \approx 211.1.$$

For $F < F^*$ the unique SPE is: firm 1 invests; in each subgame both firms play the Cournot quantities of that subgame. For $F > F^*$ the unique SPE has no investment; at $F = F^*$ both choices are optimal and there are two SPE.

Decomposition (deepening). If firm 2 kept $q_2 = 30$, firm 1's best reply with cost 5 would be $\frac{95-30}{2} = 32.5$, profit $32.5^2 = \frac{4225}{4}$: a *direct* gain of $\frac{4225}{4} - 900 = \frac{625}{4} = 156.25$. Firm 2 in fact cuts its output to $\frac{85}{3}$, adding a *strategic* gain of $\frac{10000}{9} - \frac{4225}{4} = \frac{1975}{36} \approx 54.9$. So $F^* = \frac{625}{4} + \frac{1975}{36} = \frac{1900}{9}$: about a quarter of the value of the investment comes from its effect on the rival.

(b) A Nash equilibrium that is not subgame perfect, for $F > F^*$**.** A strategy of firm 1 is (invest?, q_1^N, q_1^I); of firm 2, (q_2^N, q_2^I). Consider

$$\text{firm 1: } (I,\ q_1^N = 0,\ q_1^I = \tfrac{100}{3}), \qquad \text{firm 2: } (q_2^N = 90,\ q_2^I = \tfrac{85}{3}).$$

Firm 2 threatens to flood the market if firm 1 does not invest.

- *Firm 2:* on the path (I) it plays its best reply $\frac{85}{3}$ to $\frac{100}{3}$; its plan after N is never reached, so changing it does not change its payoff.
- *Firm 1:* on the path it earns $\frac{10000}{9} - F$. If it deviates to N, it faces $q_2 = 90$, and its best reply earns $\max_{q_1}(100 - 90 - q_1 - 10)q_1 = \max_{q_1}(-q_1^2) = 0$. So the profile is a Nash equilibrium whenever $\frac{10000}{9} - F \geq 0$.

For $F \in (F^*, \frac{10000}{9}]$ this is a Nash equilibrium in which firm 1 invests although the SPE has no investment. It is not subgame perfect: in the subgame after N, $q_2 = 90$ is not a best reply (against $q_1 = 0$ firm 2's best reply is 45), so the flood is not credible.

For *any* $F > F^*$ (including $F > \frac{10000}{9}$) another example works: firm 1 plays $(N,\ 30,\ 0)$, firm 2 plays $(30,\ 90)$, i.e. it threatens to flood if firm 1 invests. On the path both play Cournot $(30, 30)$; deviating to I would earn at most $\max_{q_1}(100 - 90 - q_1 - 5)q_1 - F = \frac{25}{4} - F < 900$. It is Nash, but the subgame after I is played at $(0, 90)$, which is not a Nash equilibrium of that subgame.

(d) Interpretation. The investment is a sunk, observable commitment, the Brexit lesson of Lecture 4 (slide 11): it pays partly because it changes the rival's behavior (firm 2 retreats from 30 to $\frac{85}{3}$), not only because it lowers firm 1's cost. Because firm 2 cannot credibly promise to respond differently, firm 1's decision depends only on the Cournot continuation values; the non-SPE equilibria show how a non-credible threat by the *rival* could otherwise push firm 1 into, or away from, investment.

(e) Verification. `check_B_L04.py` solves both Cournot subgames with sympy (quantities, profits 900 and $\frac{10000}{9}$, $\frac{7225}{9}$), computes $F^* = \frac{1900}{9}$ and its decomposition $\frac{625}{4} + \frac{1975}{36}$, and for $F = \frac{1000}{3} \in (F^*, \frac{10000}{9}]$ checks the threat profile: firm 1's best payoff after N against $q_2 = 90$ is 0 (grid of step $\frac{1}{4}$), investing is a best reply, firm 2's on-path output is its best reply, and its best reply after N would be 45, not 90; for the second example, firm 1's best payoff after investing against $q_2 = 90$ is $\frac{25}{4}$ before F. Output line: `tadelis-8-6: F* = 1900/9; flood threat supports investing for F in (F*, 10000/9]`. Tadelis prints no answer.

(f) Common mistakes.

- Computing F^* with firm 2's output held at 30 ($F^* = 156.25$): firm 2 re-optimizes in the new subgame.
- Writing strategies without the off-path quantities: firm 2's strategy must say what it produces after I *and* after N, or no threat can even be stated.
- Giving as a "non-SPE Nash equilibrium" a profile in which a firm is not best-replying on the path (e.g. firm 1 investing when $F > \frac{10000}{9}$).

Answer. (a) $F^* = \frac{10000}{9} - 900 = \frac{1900}{9} \approx 211.1$: for $F < F^*$ the unique SPE has firm 1 investing, then $(q_1, q_2) = (\frac{100}{3}, \frac{85}{3})$; without investment the Cournot play is $(30, 30)$. (b) For $F \in (F^*, \frac{10000}{9}]$: firm 1 invests and plays $\frac{100}{3}$ (and 0 after N); firm 2 plays $\frac{85}{3}$ after I and threatens $q_2 = 90$ after N. Nash, not subgame perfect. (For any $F > F^*$: no investment, Cournot $(30, 30)$, with firm 2 threatening 90 after investment.)

Item 5 (item 35 of the combined list)**: Muñoz-Garcia Ch. 4, Exercise 11**Exam · difficulty 3/5

Source: Muñoz-Garcia Ch. 4, Exercise 11, page 180. The statement is reproduced verbatim in the list (item 35); it is not repeated here.

(a) What is asked. Stackelberg duopoly with $p = 1 - Q$ and costs $c_L \le c_F < 1$: (a) the follower's best response; (b) the SPNE strategies; (c) when both outputs are positive; (d) with $c_L = c_F = c$, total output and price, compared with monopoly, Cournot and Bertrand.

(b) Method. Backward induction in a perfect-information game with a continuum of actions: solve the follower's problem for *every* q_L, substitute into the leader's problem (Tadelis §8.3.2, "Stackelberg Competition", pp. 160–163; SPE Def. 8.3, p. 157). The Cournot benchmark is Tadelis §5.2.3, p. 87. Lecture 4: SPE definition and the backward-induction fact, slide 17; sunk, observable commitments, slide 11.

(c) Solution. **(a) Follower.** Given q_L, firm F solves $\max_{q_F \ge 0} (1 - q_L - q_F)q_F - c_F q_F$. The objective is strictly concave (second derivative -2), and the first-order condition $1 - q_L - 2q_F - c_F = 0$ gives

$$q_F(q_L) \;=\; \max\Big\{\frac{1 - c_F - q_L}{2},\; 0\Big\} \;=\; \frac{1 - c_F}{2} - \frac{q_L}{2} \quad \text{for } q_L \le 1 - c_F. \tag{2}$$

The best response falls with the leader's output (slope $-\frac{1}{2}$) and shifts down when c_F rises. For $q_L \ge 1 - c_F$ the residual demand no longer covers the follower's cost, and it produces nothing.

(b) Leader. On $q_L \le 1 - c_F$, substituting (2), the price is $1 - q_L - \frac{1 - c_F - q_L}{2} = \frac{1 + c_F}{2} - \frac{q_L}{2}$, and the leader solves

$$\max_{q_L} \Big(\frac{1 + c_F}{2} - \frac{q_L}{2} - c_L\Big) q_L \quad \Longrightarrow \quad \frac{1 + c_F}{2} - q_L - c_L = 0 \quad \Longrightarrow \quad q_L^* = \frac{1 + c_F - 2c_L}{2}.$$

The objective is strictly concave (second derivative -1). Then

$$q_F^* = \frac{1 - c_F}{2} - \frac{1 + c_F - 2c_L}{4} = \frac{1 - 3c_F + 2c_L}{4}, \qquad p^* = 1 - q_L^* - q_F^* = \frac{1 + c_F + 2c_L}{4}.$$

The SPNE is the *pair of strategies*

$$\Big(q_L^* = \tfrac{1 + c_F - 2c_L}{2},\;\; q_F(q_L) = \max\big\{\tfrac{1 - c_F - q_L}{2}, 0\big\} \text{ for every } q_L \ge 0\Big),$$

not just the two numbers: the follower's plan covers every output the leader could choose, on and off the path. Each firm's output falls with its own cost and rises with the rival's.

When the formula applies. $q_L^* \le 1 - c_F$ iff $c_L \ge \frac{3c_F - 1}{2}$, i.e. exactly when $q_F^* \ge 0$. Otherwise the leader's optimum is not interior and the follower is out of the market. For $q_L \ge 1 - c_F$ the leader is a monopolist with profit $(1 - q_L - c_L)q_L$, maximized at $\frac{1 - c_L}{2}$. Then:

- $c_L > \frac{3c_F - 1}{2}$: interior, as above, both firms produce;
- $2c_F - 1 < c_L \le \frac{3c_F - 1}{2}$: *limit output*, $q_L^* = 1 - c_F$, $q_F^* = 0$ (the leader's profit rises up to $1 - c_F$ on the left piece and falls beyond it on the right piece, because $\frac{1 - c_L}{2} < 1 - c_F$);
- $c_L \le 2c_F - 1$: *blockaded entry*, $q_L^* = \frac{1 - c_L}{2}$ (monopoly), $q_F^* = 0$.

For example $(c_L, c_F) = (\frac{1}{4}, \frac{1}{2})$ gives limit output $q_L^* = \frac{1}{2}$, and $(\frac{1}{10}, \frac{7}{10})$ gives blockaded entry $q_L^* = \frac{9}{20}$.

(c) Both outputs positive.

$$q_F^* > 0 \iff 1 - 3c_F + 2c_L > 0 \iff c_L > \frac{3c_F - 1}{2}.$$

The leader's condition $q_L^* > 0 \iff c_L < \frac{1+c_F}{2}$ never binds: with $c_L \le c_F < 1$, $q_L^* \ge \frac{1-c_F}{2} > 0$. So both firms produce iff

$$\frac{3c_F - 1}{2} < c_L \le c_F,$$

a non-empty band because $\frac{3c_F-1}{2} < c_F \iff c_F < 1$. If $c_F \le \frac{1}{3}$ the lower bound is not positive, so every $c_L \in (0, c_F]$ works. The leader's cost advantage must not be too large.

(d) Symmetric costs, $c_L = c_F = c$. The condition in (c) holds ($c > \frac{3c-1}{2}$), so the interior formulas apply:

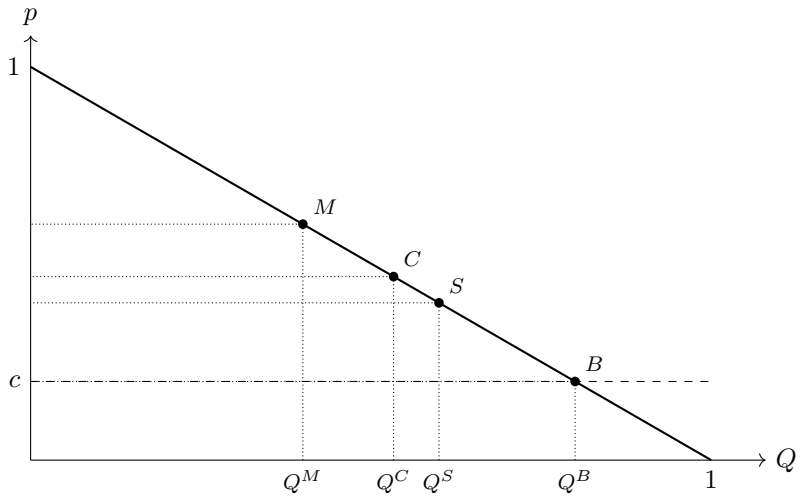
$$q_L^* = \frac{1-c}{2}, \qquad q_F^* = \frac{1-c}{4}, \qquad Q^S = \frac{3(1-c)}{4}, \qquad p^S = 1 - Q^S = \frac{1+3c}{4}.$$

The follower produces half of the leader's output. Benchmarks:

- *Monopoly*: $\max_Q (1 - Q - c)Q$ gives $Q^M = \frac{1-c}{2}$, $p^M = \frac{1+c}{2}$.
- *Cournot*: $q_i = \frac{1-c-q_j}{2}$ for both firms gives $q_i = \frac{1-c}{3}$, $Q^C = \frac{2(1-c)}{3}$, $p^C = \frac{1+2c}{3}$.
- *Bertrand* (= perfect competition with equal costs): $p^B = c$, $Q^B = 1 - c$.

Because $\frac{1}{2} < \frac{2}{3} < \frac{3}{4} < 1$,

$$Q^M < Q^C < Q^S < Q^B \qquad \text{and} \qquad p^M > p^C > p^S > p^B.$$



Demand $p = 1 - Q$ drawn for $c = 0.2$: $Q^M = 0.4$, $Q^C \approx 0.53$, $Q^S = 0.6$, $Q^B = 0.8$; prices $\frac{1+c}{2}$, $\frac{1+2c}{3}$, $\frac{1+3c}{4}$, c.

Profits: the leader earns $(p^S - c)q_L^* = \frac{(1-c)^2}{8}$, more than its Cournot profit $\frac{(1-c)^2}{9}$; the follower earns $\frac{(1-c)^2}{16}$, less.

(d) Interpretation. The leader's advantage comes from commitment, not from information: its output is observed and cannot be revised, so the follower takes it as given and contracts. The leader expands past its Cournot output ($\frac{1-c}{2} > \frac{1-c}{3}$) because it knows the follower will respond by cutting back. The follower would like to threaten "I produce $\frac{1-c}{3}$ whatever you do". Against that plan, $q_L = \frac{1-c}{3}$ is a best reply, so the Cournot outcome is a Nash equilibrium of the sequential game (Tadelis p. 162). The threat is not credible: once q_L is sunk, $\frac{1-c}{3}$ is not the follower's best

response to any $q_L \neq \frac{1-c}{3}$. Subgame perfection removes it (slide 17). The cost regimes in (b) carry the same idea: a large enough cost advantage lets the leader commit to the output $1 - c_F$, which leaves the follower no room to produce (limit output). This is the quantity version of entry deterrence through an observable, irreversible commitment (slide 11). For consumers, sequential quantity competition is more competitive than Cournot: total output rises from $\frac{2}{3}$ to $\frac{3}{4}$ of the competitive level.

(e) Verification. `check_D.py` derives BR_F, q_L^*, q_F^*, p^* and the cutoff $c_L = \frac{3c_F-1}{2}$ with sympy. For six cost pairs covering the three regimes it finds the leader's best output by exact search over $q_L \in \{k/2000\}$, using the truncated best response, and confirms the regime formula each time. It also checks the four symmetric totals, prices and profits, the strict ranking at four values of c, and that the Cournot outcome is a Nash but not a subgame-perfect profile. Output: "Munoz-Garcia 4.11: BR_F = (1-cF-qL)/2; qL* = (1+cF-2cL)/2, qF* = (1-3cF+2cL)/4; both > 0 iff cL > (3cF-1)/2; Q: monopoly (1-c)/2 < Cournot 2(1-c)/3 < Stackelberg 3(1-c)/4 < Bertrand 1-c (book: same)". *Comparison with Muñoz-Garcia's answer* (pp. 181–185): the same best response, SPNE, cutoffs (the book's C_A: $c_L > -\frac{1}{2} + \frac{3}{2}c_F$; C_B: $c_L < \frac{1}{2} + \frac{1}{2}c_F$, not binding below the 45° line), symmetric quantities and the ranking in its Fig. 4.34. The book omits the truncation at $q_F = 0$, so its best response and SPNE formula are valid only inside the region of part (c). The limit-output and blockaded regimes above are not in the book.

(f) Common mistakes.

- Reporting the SPNE as the numbers (q_L^*, q_F^*). The follower's strategy is the whole function $q_F(q_L)$; the numbers are the equilibrium outcome.
- Solving both first-order conditions simultaneously, which gives the Cournot quantities $\frac{1-c}{3}$, instead of substituting the follower's response into the leader's profit first.
- Using the interior formula $q_F^* = \frac{1-3c_F+2c_L}{4}$ when it is negative. When it is, the follower produces zero and the leader's problem changes.

Answer. (a) $q_F(q_L) = \max\{\frac{1-c_F-q_L}{2}, 0\}$. (b) SPNE: $q_L^* = \frac{1+c_F-2c_L}{2}$ and the follower plays $q_F(q_L)$ after every q_L; outcome $q_F^* = \frac{1-3c_F+2c_L}{4}$, $p^* = \frac{1+c_F+2c_L}{4}$ (valid when $q_F^* \geq 0$). (c) Both produce iff $c_L > \frac{3c_F-1}{2}$; $q_L^* > 0$ always. (d) $Q^S = \frac{3(1-c)}{4}$, $p^S = \frac{1+3c}{4}$; $Q^M = \frac{1-c}{2} < Q^C = \frac{2(1-c)}{3} < Q^S < Q^B = 1 - c$, and prices in the reverse order (book: same).

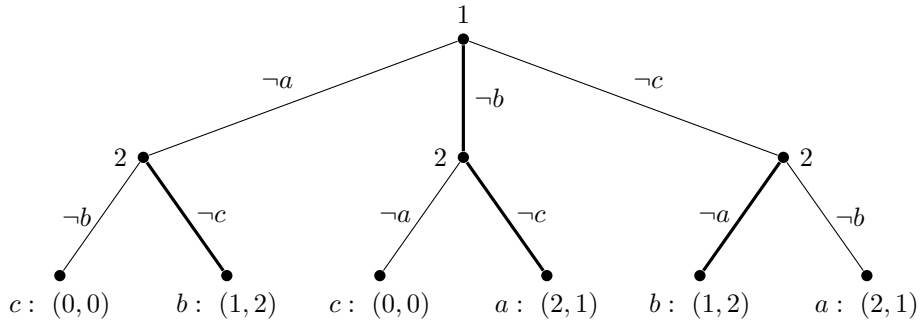
Item 6 (item 36 of the combined list)**: Tadelis Exercise 8.11** Exam · difficulty 3/5

Source: Tadelis Exercise 8.11, page 172–173. The statement is reproduced verbatim in the list (item 36); it is not repeated here.

(a) What is asked. (a) The game tree of the two-veto procedure with payoffs 2/1/0; (b) the SPE and whether it is unique; (c) Nash equilibria that are not subgame perfect; (d) whether player 2 would use a pre-game e-mail to remove a candidate, and which one.

(b) Method. Backward induction in a finite game of perfect information (Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 152–153). Every decision node starts a subgame, so the SPE are exactly the backward-induction profiles (Def. 8.3, p. 157, and the Fact on p. 158). Nash equilibria through the normal form of the tree (§7.2.3, p. 143) and the on/off-path distinction (Def. 7.8, p. 146). Lecture 4: non-credible threats, slides 8 and 13; subgames, slide 14; SPE, slide 17.

(c) Solution. **(a) Tree.** Payoffs: $u_1(a,b,c) = (2,1,0)$ and $u_2(a,b,c) = (1,2,0)$. Player 1 vetoes one candidate, player 2 vetoes one of the two left, and the last one is assigned. "$\neg j$" means "veto j"; leaves show the assigned candidate and (u_1, u_2).



Thick branches = backward-induction choices.

Player 1 has 3 pure strategies. Player 2 has $2^3 = 8$: write $s_2 = (x_a, x_b, x_c)$, where x_j is the candidate she vetoes after player 1 vetoed j.

(b) Backward induction. Player 2's three nodes:

- after $\neg a$ ($\{b, c\}$ left): $\neg c$ leaves b, $u_2 = 2 > 0$, so $x_a = c$;
- after $\neg b$ ($\{a, c\}$ left): $\neg c$ leaves a, $u_2 = 1 > 0$, so $x_b = c$;
- after $\neg c$ ($\{a, b\}$ left): $\neg a$ leaves b, $u_2 = 2 > 1$, so $x_c = a$.

Player 1 anticipates the outcomes b, a, b of $\neg a$, $\neg b$, $\neg c$, worth $1, 2, 1$ to him, and vetoes b. The SPE is

$$\big(\ \neg b,\ \ s_2 = (c,\ c,\ a)\ \big), \qquad \text{outcome } a, \quad (u_1, u_2) = (2, 1).$$

It is unique: every comparison above is strict, so backward induction never has to break a tie. Player 1 vetoes the rival's favourite, b, not his own least preferred, c. Player 2 will veto c herself, so a veto spent on c is wasted.

(c) Nash equilibria that are not subgame perfect. Enumerating the $3 \times 8 = 24$ pure profiles gives 7 Nash equilibria:

Player 1	Player 2 (x_a, x_b, x_c)	outcome, (u_1, u_2)	SPE?
$\neg b$	(c, c, a)	a, $(2, 1)$	yes
$\neg b$	(b, c, a), (c, c, b), (b, c, b)	a, $(2, 1)$	no
$\neg a$	(c, a, a)	b, $(1, 2)$	no
$\neg c$	(c, a, a), (b, a, a)	b, $(1, 2)$	no

So yes, there are six. Two kinds:

- *Same outcome, different off-path plans.* Player 1 vetoes b and player 2 vetoes c on the path. Off the path she plans to veto b after $\neg a$ (leaving c: 0 instead of 2) or b after $\neg c$ (leaving a: 1 instead of 2). These nodes are never reached, so the plans cost her nothing, but they are not best responses in the subgames that start there.
- *A different outcome, supported by a threat.* With $x_b = a$, player 2 announces: "if you veto b, I veto a, and c gets the job." Facing it, $\neg b$ gives player 1 0, so he vetoes a or c and player 2 gets her favourite b. The threat is not credible: after $\neg b$ her choice is between a (1) and c (0), and she would veto c. Check of one of them, $(\neg c,\ (c, a, a))$: player 1 gets 1 against 0 from $\neg b$ and 1 from $\neg a$; player 2 gets 2, her maximum. The threatened node is off the path, so the threat is never tested.

No Nash equilibrium elects c: on the path, player 2 could always veto c instead and gain.

(d) The alienating e-mail. Read as follows: the e-mail comes first and its effect is seen by both. Then player 1 vetoes one of the two remaining candidates, and the last one is assigned (player 2's veto has nothing left to act on). In each two-candidate subgame player 1 keeps his preferred candidate:

Player 2's first move	candidates left	player 1 vetoes	outcome, (u_1, u_2)
no e-mail	$\{a, b, c\}$	b (part (b))	a, $(2, 1)$
e-mail a	$\{b, c\}$	c	b, $(1, 2)$
e-mail b	$\{a, c\}$	c	a, $(2, 1)$
e-mail c	$\{a, b\}$	b	a, $(2, 1)$

Player 2 gets 2 only by e-mailing a, so she does. In the SPE of the extended game, she e-mails a, player 1 vetoes c, and b is assigned, $(u_1, u_2) = (1, 2)$. The strategy profile also specifies the original SPE after "no e-mail" and player 1's vetoes after the other two e-mails.

(d) Interpretation. The order of moves decides the outcome. Moving first, player 1 removes the rival's favourite. Player 2 then cannot hurt him, because the only threat she has ("I'll leave you c") hurts her too. The e-mail works where the threat fails because it is an irreversible move made before player 1 acts, not a promise about later behaviour. Player 2 removes a, her own second choice. That looks self-defeating, but it takes away player 1's best option, and with only $\{b, c\}$ left he prefers b. A smaller choice set helps the player who controls it. This is the same commitment logic as the Brexit preparation (slide 11): an observable, sunk action changes what the opponent can do, while announcing future vetoes changes nothing ("talk is cheap", slide 11; non-credible threats, slide 13).

(e) Verification. `check_D.py` builds the tree, runs backward induction (asserting no ties), enumerates all 24 pure profiles and tests each one for Nash and for subgame perfection by definition (Nash in every subgame). It then builds the extended game with the e-mail move and

solves it backward. Output: "Tadelis 8.11: SPE veto b, then (bc: veto c, ac: veto c, ab: veto a), outcome a (2,1), unique; 7 pure NE (4 with outcome a, 3 with outcome b), 6 not SPE" and "Tadelis 8.11 (d): player 2 e-mails candidate a; player 1 then vetoes c; outcome b (1,2)". (In the script, node "bc" is player 2's node after $\neg a$, and so on.) Tadelis prints no answer.

(f) Common mistakes.

- Having player 1 veto his least-preferred candidate c. Player 2 would veto c anyway; the useful veto removes her favourite.
- Writing player 2's strategy as one veto. She has three information sets, so a strategy is a triple, and it is the off-path entries that separate the Nash equilibria from the SPE.
- In (d), removing c or b ("remove the worst" or "remove the rival's favourite" by analogy). Only removing a changes player 1's choice.

Answer. (a) Tree above; player 1 has 3 strategies, player 2 has 8. (b) Unique SPE: player 1 vetoes b; player 2 vetoes c after $\neg a$, c after $\neg b$, a after $\neg c$; a is selected, payoffs $(2, 1)$. (c) Yes: 7 pure Nash equilibria, 6 not subgame perfect. Three of them elect b, payoffs $(1, 2)$, using player 2's non-credible threat to veto a after $\neg b$. (d) Yes: she e-mails a; player 1 then vetoes c and b is selected, payoffs $(1, 2)$.

Item 7 (item 37 of the combined list)**: Tadelis Exercise 8.4** Exam · difficulty 4/5

Source: Tadelis Exercise 8.4, page 170. The statement is reproduced verbatim in the list (item 37); it is not repeated here.

(a) What is asked. Leader firm 1 chooses q_1; firms 2 and 3 observe it and choose q_2, q_3 simultaneously; $P = a - Q$, marginal cost c. (a) Number of proper subgames; (b) perfect or imperfect information; (c) the SPE and its uniqueness; (d) a Nash equilibrium that is not subgame perfect.

(b) Method. Proper subgames and perfect information (Lecture 4, slides 14–16; Lecture 3, slide 24; Tadelis Defs. 7.3 and 8.2, pp. 136 and 154). Two-stage game with a simultaneous second stage (Gibbons §2.2.A, pp. 71–73): for each q_1 solve the followers' Cournot game, then let firm 1 optimize (Stackelberg logic, Lecture 3, slide 37; Tadelis §8.3.2, pp. 160–163). Non-SPE Nash equilibria by off-path threats (Lecture 4, slides 8 and 13).

(c) Solution. Notation. $a > c \geq 0$; $q_i \geq 0$; profit $\pi_i = (a - Q - c)\, q_i$ with $Q = q_1 + q_2 + q_3$. A strategy of follower $j \in \{2, 3\}$ is a function $q_j(\cdot)$ of the observed q_1.

(a) Proper subgames. After each $q_1 \geq 0$ the game reaches a node at which firm 2 moves; that node is a singleton information set (everyone knows q_1), and the simultaneous choice of firm 3 lies in an information set contained in what follows it. So every q_1 starts a proper subgame: there is a **continuum** of them (one for each $q_1 \in [0, \infty)$), plus the whole game. No other node starts one: firm 3's nodes belong to an information set that pools all of firm 2's quantities.

(b) Imperfect information. Firms 2 and 3 move simultaneously, so one of them (say firm 3) does not observe the other's choice: its information set contains many nodes. The game has **imperfect information** (complete information, but not perfect).

(c) SPE and uniqueness.

Step 1: the subgame after q_1. If $q_1 < a - c$, the followers' first-order conditions are

$$\frac{\partial \pi_2}{\partial q_2} = a - c - q_1 - 2q_2 - q_3 = 0, \tag{3}$$

$$\frac{\partial \pi_3}{\partial q_3} = a - c - q_1 - q_2 - 2q_3 = 0. \tag{4}$$

Subtracting (4) from (3) gives $q_3 = q_2$; substituting, $a - c - q_1 - 3q_2 = 0$, so

$$q_2(q_1) = q_3(q_1) = \frac{a - c - q_1}{3}.$$

Uniqueness in the subgame. Each profit is strictly concave in the firm's own output, so best replies are $q_j = \max\{0, \frac{a-c-q_1-q_k}{2}\}$. The system (3)–(4) has determinant $2 \cdot 2 - 1 = 3 \neq 0$, so the interior solution is unique; a corner with $q_2 = 0 < q_3$ would need $q_3 = \frac{a-c-q_1}{2}$ and $0 \geq \frac{a-c-q_1-q_3}{2} = \frac{a-c-q_1}{4} > 0$, impossible. The best-reply map has slope $-\frac{1}{2}$, a contraction, so there is no other (also no mixed) equilibrium: the subgame Cournot game is dominance solvable. If $q_1 \geq a - c$, both followers' unique best reply is 0.

Step 2: the leader. For $q_1 < a - c$,

$$\pi_1 = \Big(a - q_1 - 2 \cdot \frac{a - c - q_1}{3} - c\Big) q_1 = \frac{a - c - q_1}{3}\, q_1,$$

strictly concave, with

$$\frac{d\pi_1}{dq_1} = \frac{a - c - 2q_1}{3} = 0 \;\Rightarrow\; q_1^* = \frac{a - c}{2}.$$

For $q_1 \geq a - c$ the price is at most c, so $\pi_1 \leq 0$; the interior choice is the unique global maximum.

Step 3: the SPE.

$$q_1^* = \frac{a-c}{2}, \qquad q_2^*(q_1) = q_3^*(q_1) = \max\Big\{0,\ \frac{a-c-q_1}{3}\Big\}.$$

On the path $q_2 = q_3 = \frac{a-c}{6}$, $Q^* = \frac{5(a-c)}{6}$, $P^* = \frac{a+5c}{6}$, $\pi_1^* = \frac{(a-c)^2}{12}$, $\pi_2^* = \pi_3^* = \frac{(a-c)^2}{36}$. *Uniqueness:* every subgame has a unique Nash equilibrium (Step 1), so in any SPE the followers must use exactly these functions; given them, firm 1's problem has a unique solution (Step 2).

(d) A Nash equilibrium that is not subgame perfect. Let $\hat{q} = \frac{a-c}{4}$ and consider

$$q_1 = \hat{q}, \qquad q_2(q_1) = q_3(q_1) = \begin{cases} \hat{q}, & q_1 = \hat{q}, \\ a - c, & q_1 \neq \hat{q}. \end{cases}$$

- *Followers:* on the path $q_1 = \hat{q}$, each best reply to $(\hat{q}, \hat{q})$ is $\frac{a-c-2\hat{q}}{2} = \frac{a-c}{4} = \hat{q}$. Their plans after other q_1 are off the path and do not affect their payoffs.
- *Leader:* on the path the outcome is three-firm Cournot, $\pi_1 = \frac{(a-c)^2}{16} > 0$. Any $q_1 \neq \hat{q}$ triggers $q_2 = q_3 = a - c$, so $Q \geq 2(a-c)$ and $P - c \leq -(a-c) < 0$: profit ≤ 0. So $\hat{q}$ is a best reply.

It is a Nash equilibrium with outcome $(\frac{a-c}{4}, \frac{a-c}{4}, \frac{a-c}{4})$, not the SPE outcome. It is not subgame perfect: after any $q_1 \neq \hat{q}$ (with $q_1 < a - c$) the pair $(a - c, a - c)$ is not a Nash equilibrium of the subgame (each follower's best reply to $a - c$ is 0).

(d) Interpretation. Moving first and being observed is a commitment: firm 1 produces the monopoly output $\frac{a-c}{2}$ and the followers' joint reply (slope $-\frac{2}{3}$) absorbs part of it, so the leader earns $\frac{(a-c)^2}{12}$ instead of the Cournot $\frac{(a-c)^2}{16}$, and total output rises. The non-SPE equilibrium in (d) shows how the followers could "undo" leadership with a threat to flood the market, but the threat is not credible once q_1 is sunk: this is the Lecture 4 contrast between Nash equilibria held up by threats and the subgame-perfect prediction.

(e) Verification. `check_B_L04.py` solves (3)–(4) and the leader's problem with sympy (determinant 3; $q_1^* = \frac{a-c}{2}$, $q_j^* = \frac{a-c}{6}$, second derivative negative); with $a = 13$, $c = 1$ it iterates the followers' best replies from three starting points for six values of q_1 (including $q_1 \geq a - c$) and always converges to $\max\{0, \frac{a-c-q_1}{3}\}$; it checks the threat profile of (d): on-path profit 9, every deviation on a grid of step $\frac{1}{4}$ earns ≤ 0, the followers' on-path reply is 3, and flooding is not a best reply. Output line: `tadelis-8-4: SPE q1 = (a-c)/2, q2 = q3 = (a-c)/6; flood-threat NE gives Cournot (a-c)/4 each`. Tadelis prints no answer. The SPE outcome agrees with the 01-04 solution of Gibbons 2.6 (same game).

(f) Common mistakes.

- Counting "two" subgames (one per stage): every value of q_1 starts its own subgame.
- Giving the followers' equilibrium *quantities* $\frac{a-c}{6}$ as their strategies: an SPE needs the reply functions $q_j(q_1)$.
- Solving the followers' problem as a Stackelberg chain (firm 2 before firm 3): here they move simultaneously, so the second stage is a Cournot duopoly on residual demand $a - q_1$.

Answer. (a) A continuum: one proper subgame for each $q_1 \geq 0$, plus the whole game. (b) Imperfect information (firms 2 and 3 move simultaneously). (c) Unique SPE: $q_1^* = \frac{a-c}{2}$, $q_2(q_1) = q_3(q_1) = \max\{0, \frac{a-c-q_1}{3}\}$; outcome $(\frac{a-c}{2}, \frac{a-c}{6}, \frac{a-c}{6})$, $P = \frac{a+5c}{6}$. (d) $q_1 = \frac{a-c}{4}$; followers play $\frac{a-c}{4}$ if $q_1 = \frac{a-c}{4}$ and $a - c$ otherwise: Nash, not subgame perfect.

Item 8 (item 38 of the combined list)**: Tadelis Exercise 8.13** Exam · difficulty 4/5

Source: Tadelis Exercise 8.13, page 173–174. The statement is reproduced verbatim in the list (item 38); it is not repeated here.

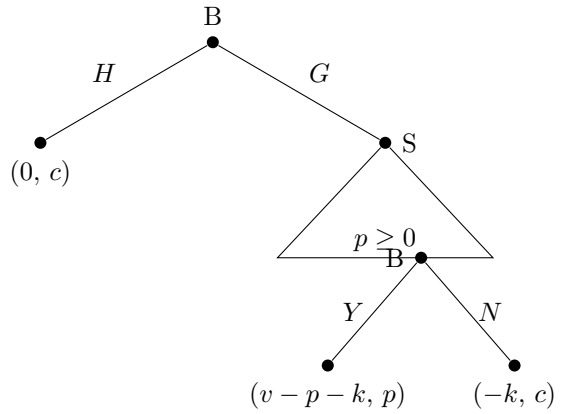
(a) What is asked. A buyer pays a commuting cost k to reach a seller, who then names a take-it-or-leave-it price. (a) Extensive form and the buyer's reply at the acceptance node; (b) the SPE, its uniqueness and Pareto optimality; (c) Nash equilibria better for both; (d) whether the seller would pay ε to mail a postcard that commits him to a price.

(b) Method. Backward induction with a continuum of actions (Lecture 3, slides 30 and 37; Tadelis §8.1 and §8.3.2, the Stackelberg logic, pp. 160–163); the ultimatum-game argument for ties (Tadelis Prop. 11.2, p. 223: the responder must accept at indifference, or the proposer has no best offer). Nash equilibria sustained by off-path threats (Lecture 4, slides 8 and 13). Commitment as a move added before the game (Lecture 4, slide 11; `Notas/L03/L03-06`, Cabral §7.2).

(c) Solution. Notation. Buyer (B) and seller (S). B chooses G (go to the store) or H (stay home); then S names $p \geq 0$; then B accepts (Y) or walks out (N). Data: $c > 0$, $k > 0$, $v > c + k$. Payoffs (u_B, u_S):

$$H:\ (0,\ c); \qquad G, p, Y:\ (v - p - k,\ p); \qquad G, p, N:\ (-k,\ c).$$

(a) Extensive form and the buyer's reply.



The triangle stands for the continuum of prices; one typical price p is drawn.

At the node after price p, the commuting cost is already sunk; it appears in both payoffs. B accepts if

$$v - p - k \ \geq\ -k \iff p \leq v.$$

So the best reply is: accept every $p < v$, reject every $p > v$, and either action at $p = v$.

(b) SPE, uniqueness, efficiency.

Step 1: the acceptance rule at $p = v$. Suppose B rejected $p = v$. Then S would get p for every $p < v$ and $c < v$ otherwise, so his payoff approaches v as $p \uparrow v$ but never reaches it; S would have no best offer, so the profile cannot be an SPE. Hence in every SPE B accepts if and only if $p \leq v$.

Step 2: the seller. Given that rule, S's payoff is p for $p \leq v$ and $c < v$ for $p > v$; the unique maximizer is $p^* = v$.

Step 3: the buyer's commuting decision. Going yields

$$u_B(G) = v - p^* - k = v - v - k = -k \;<\; 0 = u_B(H),$$

so B stays home. Every step had a unique optimum, so the SPE is unique:

$$\text{B: } H\text{, and accept iff } p \le v; \qquad \text{S: offer } p = v. \qquad \text{Outcome: no trade, payoffs } (0,\ c).$$

Not Pareto optimal. Trade at any price $p \in (c,\ v-k)$ (nonempty since $v > c+k$) gives $u_B = v - p - k > 0$ and $u_S = p > c$: both strictly better. The lost surplus is $v - c - k > 0$. This is a hold-up: the commuting cost is sunk when the price is set, so S extracts all of v and B, anticipating it, never makes the trip.

(c) Better Nash equilibria. Yes. Fix $\hat{p} \in (c,\ v-k)$ and consider

$$\text{B: } G\text{, accept iff } p \le \hat{p}; \qquad \text{S: offer } \hat{p}.$$

S's best reply: offers $p \le \hat{p}$ are accepted and give $p \le \hat{p}$; offers above are rejected and give $c < \hat{p}$. So $\hat{p}$ is optimal. *B's best reply:* given $\hat{p}$, going and accepting gives $v - \hat{p} - k > 0$, more than staying (0) or going and rejecting $(-k)$; B's rule at prices other than $\hat{p}$ is off the path and does not affect his payoff. Payoffs $(v - \hat{p} - k,\ \hat{p})$ beat $(0, c)$ for both. The same construction is a Nash equilibrium for every $\hat{p} \in [c, v-k]$ (the endpoints give one player only a weak gain), so the set of Nash payoffs that Pareto-dominate the SPE is $\{(v - \hat{p} - k, \hat{p}) : \hat{p} \in (c, v-k)\}$. None is subgame perfect: after an off-path offer $p \in (\hat{p}, v)$, rejecting gives $-k < v - p - k$, so the threat to walk out is not credible.

(d) The postcard. Add a first move: S either mails a postcard committing to $\hat{p}$ (at cost ε) or does not. Without the postcard the game is the one in (b): payoff c to S. With it, B buys at $\hat{p}$ if he goes and $\hat{p} \le v$, so he goes iff $v - \hat{p} - k \ge 0$, i.e. $\hat{p} \le v - k$. By the argument of Step 1, B goes at indifference. Then S's payoff from the postcard is

$$u_S(\hat{p}) = \begin{cases} \hat{p} - \varepsilon, & \hat{p} \le v - k, \\ c - \varepsilon, & \hat{p} > v - k, \end{cases} \qquad \Rightarrow \qquad \hat{p}^* = v - k, \quad u_S = v - k - \varepsilon.$$

Compared with not mailing,

$$(v - k - \varepsilon) - c = (v - c - k) - \varepsilon > 0 \quad \text{since } \varepsilon < v - c - k.$$

So in the SPE the seller mails the postcard with price $v-k$; B goes and buys; payoffs $(0,\ v-k-\varepsilon)$. Trade now happens; the total surplus is $v - c - k - \varepsilon$, all of it to the seller.

(d) Interpretation. The commitment in (c) has to come from the buyer and is not credible; the commitment in (d) comes from the seller, who limits his own future options, and it is credible because the coupon binds him. These are the three Lecture 4 lessons (slide 11) in one example: the buyer's *sunk* commuting cost is what exposes him to hold-up; "I will only pay $\hat{p}$" is *cheap talk*; and the postcard works because it is *observable* before the buyer decides and changes what the seller can do afterwards. "Junk mail" has value even when it contains no information. The seller still captures the whole surplus; the coupon only removes the inefficiency, the same logic as a bond in the entry game (Cabral §7.2).

(e) Verification. `check_B_L04.py` uses $c = 2$, $v = 10$, $k = 1$, $\varepsilon = \frac{1}{2}$ and prices on a 0.25 grid: the seller's best offer is $p = 10 = v$ and the buyer stays home; under "reject at v" the best grid offer is $v - 0.25$ (no maximum on a continuum); thresholds $\hat{p} \in \{2.5, 5, 8.5\}$ give Nash equilibria that beat $(0, c)$ for both; with the postcard, $\hat{p}^* = 9 = v - k$ and $8.5 > 2$; sympy confirms the gain $(v - c - k) - \varepsilon$. Output line: `tadelis-8-13: SPE p = v, no trade; postcard at v - k is bought, seller nets v - k - eps`. Tadelis prints no answer.

(f) Common mistakes.

- Letting the buyer reject at $p = v$ in the SPE: then the seller has no optimal offer, so no SPE exists with that rule.
- Including k in the acceptance decision ("accept iff $p \leq v - k$"): k is sunk at that node.
- Pricing the postcard at v: the buyer would then not commute; the coupon must leave him $v - \hat{p} - k \geq 0$.

Answer. (a) Tree above; at the acceptance node the buyer accepts iff $p \leq v$ (the sunk k does not matter). (b) Unique SPE: seller offers $p = v$, buyer accepts iff $p \leq v$ and therefore stays home; payoffs $(0, c)$; not Pareto optimal, since trade at any $p \in (c, v - k)$ helps both. (c) Yes: buyer goes and accepts only $p \leq \hat{p}$, seller offers $\hat{p}$, for any $\hat{p} \in (c, v - k)$; Nash but not subgame perfect. (d) Yes: the seller mails a coupon at $\hat{p} = v - k$, the buyer comes and buys, seller earns $v - k - \varepsilon > c$.

Item 9 (item 39 of the combined list)**: Tadelis Exercise 8.7** Advanced · difficulty 4/5

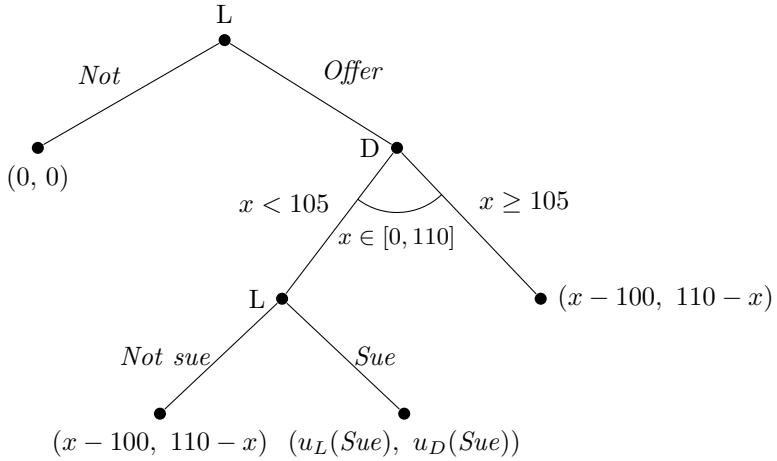
Source: Tadelis Exercise 8.7, page 171. The statement is reproduced verbatim in the list (item 39); it is not repeated here.

(a) What is asked. For the loan game with no courts, free courts and costly courts: the trees and SPE, with uniqueness ((a), (b), (d)); Nash equilibria that are not SPE ((c), (e)); what the lender would pay for a "loser pays the legal fees" law (f); and whether a planner should adopt it (g).

(b) Method. Backward induction with a continuum of actions: for every repayment x, solve the lender's suit decision; then the debtor's choice of x; then the lender's offer (Tadelis §8.1, Def. 8.1, p. 152; SPE Def. 8.3, p. 157; the Fact on p. 158). Nash equilibria that are not SPE rely on plans at nodes off the equilibrium path (Def. 7.8, p. 146; §8.2, pp. 157–158). Lecture 4: credible and non-credible threats, slides 8, 11 and 13; SPE, slide 17.

(c) Solution. Set-up and reading of the statement. Player 1 (lender, L) chooses *Offer* or *Not*. After *Offer*, player 2 (debtor, D) repays $x \in [0, 110]$. Repaying a negative amount is not an option, so $x \geq 0$. Without a suit, $(u_L, u_D) = (x - 100,\ 110 - x)$. When courts exist and $x < 105$, the lender then chooses *Sue* or *Not sue*. A suit brings the total repayment to 105, with the x already paid counting toward it. The debtor then pays the fine or the fee on top. The three legal regimes differ only in the payoffs after *Sue*:

	(b) free courts, fine 5	(d) lender pays fee 105	(f) debtor pays fee 105
$u_L(Sue)$	$105 - 100 = 5$	$105 - 100 - 105 = -100$	$105 - 100 = 5$
$u_D(Sue)$	$110 - 105 - 5 = 0$	$110 - 105 - 5 = 0$	$110 - 105 - 105 = -100$



Payoffs (u_L, u_D). The arc marks the continuum of repayments x. The lender's node after $x < 105$ exists only in (b), (d) and (f); in (a) every x ends the game with $(x - 100, 110 - x)$.

A pure strategy of the lender is an offer decision plus a suit rule $sue(x)$ for every $x < 105$: a function, not a single choice. A pure strategy of the debtor is a number x.

(a) No legal system. *Debtor:* $110 - x$ is strictly decreasing, so $x^* = 0$, giving $(u_L, u_D) = (-100, 110)$. *Lender: Not* gives $0 > -100$. SPE: (*Not*; $x = 0$), payoffs $(0, 0)$. It is unique, since both maximizers are unique.

(b) Costless courts. *Lender after a default $x < 105$: Sue* gives $5 > x - 100$, so he sues after every default. *Debtor:* any $x < 105$ is followed by a suit and gives 0; $x \in [105, 110]$ gives $110 - x \leq 5$, with the maximum at $x = 105$. Since $5 > 0$, $x^* = 105$. *Lender at the root: Offer* gives $105 - 100 = 5 > 0$. SPE:

$$(\textit{Offer},\ \textit{sue}(x) = \textit{Sue}\ \forall x < 105;\quad x = 105), \qquad (u_L, u_D) = (5, 5).$$

It is unique: every comparison is strict and $x = 105$ is the only maximizer.

(c) Nash equilibria of (b) that are not SPE: none in pure strategies.

- *No pure NE without lending.* Whatever x the debtor plans, the lender can switch to *Offer* and to a suit rule that is best at that x. He then gets 5 if $x < 105$ (sue) or $x - 100 \geq 5$ if $x \geq 105$, more than 0. So *Not* is never a best reply. Unlike a threat against the opponent, a lender's plan "I would not sue" cannot keep him out: he controls it.
- *With lending, the profile is forced.* If $x > 105$ the debtor gains by moving to 105. If $x < 105$ the lender gains by suing (if he does not) or the debtor gains by moving to 105 (if he does: $5 > 0$). So $x = 105$. If the suit rule skipped even one $x' < 105$, the debtor would move there and get $110 - x' > 5$. So the rule must be "sue after every $x' < 105$", which is the SPE.

So in pure strategies the only Nash equilibrium is the SPE. With behavioural strategies the answer changes, though not the outcome. The lender may randomize at unreached defaults and sue after x' with probability $s(x')$, provided the debtor still prefers 105: $(1 - s(x'))(110 - x') \leq 5$, i.e. $s(x') \geq 1 - \frac{5}{110 - x'}$ (e.g. $\frac{1}{6}$ at $x' = 104$). Any $s(x') < 1$ is not sequentially rational, since suing is strictly better. These are Nash equilibria that are not SPE, with the same path and payoffs $(5, 5)$.

(d) Costly courts (lender pays the fee). *Lender after a default $x < 105$: Sue* gives -100 and *Not sue* gives $x - 100$. Not suing is strictly better for every $x > 0$, and at $x = 0$ the two tie ($-100 = -100$). *Debtor:* if the lender does not sue at $x = 0$, the debtor's payoff $110 - x$ on $[0, 105)$ is maximized at $x = 0$, giving 110, which beats $110 - x \leq 5$ on $[105, 110]$. Could the lender instead sue at $x = 0$, with probability $s > 0$? The debtor would get $110(1 - s) < 110$ at $x = 0$ and $110 - x$ at every small $x > 0$. The supremum 110 is then not attained, so the debtor has no best reply. The tie at 0 must therefore be broken toward *Not sue*. *Lender at the root: Offer* gives $-100 < 0$. The unique SPE is

$$(\textit{Not},\ \textit{sue}(x) = \textit{Not sue}\ \forall x < 105;\quad x = 0), \qquad (u_L, u_D) = (0, 0),$$

the outcome of (a). A legal system that is too expensive to use is the same as none.

(e) Nash equilibria of (d) that are not SPE: yes.

- *Lending supported by a non-credible threat:*

  $$(\textit{Offer},\ \textit{sue}(x) = \textit{Sue}\ \forall x < 105;\quad x = 105), \qquad (u_L, u_D) = (5, 5).$$

  It is a NE. The debtor gets 5 at $x = 105$ and 0 at any default, since he is sued. The lender gets $5 > 0$, and no suit node is reached, so the suit rule costs him nothing. It is not an SPE. In the subgame after any default $x > 0$, suing gives $-100 < x - 100$. As in (c), this is the only NE with lending in pure strategies: one gap in the suit rule invites the debtor to default there. It is better for *both* players than the SPE $(0, 0)$. The threat is what would make lending possible, but it is not credible.
- *No lending, with other plans: Not*, any planned $x \leq 100$ and any suit decision at that x. Offering would give the lender at most $\max\{-100,\ x - 100\} = x - 100 \leq 0$. All of these except the SPE are Nash equilibria that are not subgame perfect, e.g. (*Not*, never sue; $x = 50$), because $x = 50$ is not the debtor's best reply after *Offer*.

(f) Loser pays the fees. After a default $x < 105$, *Sue* gives the lender $105 - 100 = 5 > x - 100$: the suit is now credible. The debtor gets -100 from any default and $110 - x$ from $x \in [105, 110]$,

so $x = 105$ ($5 > -100$). Offering gives $5 > 0$. Unique SPE: (*Offer*, *Sue* $\forall x < 105$; $x = 105$), payoffs $(5, 5)$. Compared with (d) the lender's equilibrium payoff rises from 0 to 5. Anything he pays for the reform is sunk before the game starts and does not change the SPE, so he gains iff the price is below \$5. He would pay *up to \$5*.

(g) The planner. Total surplus is 0 without the law (no loan) and $110 - 100 = 10$ with it. On the equilibrium path nobody defaults, so nobody is sued and the \$105 fee is never paid. The reform moves both players from $(0, 0)$ to $(5, 5)$, a Pareto improvement, so the planner should adopt it if enacting it costs less than \$10 (the debtor too would pay up to \$5). The harsh off-path penalty (-100 for the debtor) has no social cost in this model because it is never imposed. It would have one if default could happen by bad luck, which the model rules out.

(d) Interpretation. The project creates \$10 of value, but the lender can capture it only if the debtor believes default will be punished. In (a) and (d) the debtor's promise to repay is not credible, and in (d) neither is the lender's threat to sue, because a suit costs more than it recovers. Lending collapses even though both would gain. The Nash equilibrium in (e) shows the gains a credible threat would deliver. The SPE shows why the threat by itself does not work. The reform in (f) does not change what happens on the path; it changes who pays for enforcement *off* the path. That makes suing the lender's best reply to default, so the threat becomes credible and is never used. This is the Brexit lesson in a credit market (slides 11 and 13): the institution that makes a threat credible creates value, while announcing the threat creates none. Free courts in (b) and loser-pays in (f) give the same outcome.

(e) Verification. `check_D.py` encodes the payoffs of the four regimes exactly and works on the repayment grid $x \in \{k/8\}$, which contains 0, 100, 104.5 and 105. It computes the sequentially rational suit choice at every default, showing the tie at $x = 0$ in (d) and strict preferences elsewhere. It then computes the debtor's best x and the lender's offer. It tests the claimed Nash equilibria with unilateral deviations: the lender's joint offer/suit deviations and every debtor x. It confirms that a single gap in the suit rule breaks the threat equilibria and that no no-lending NE exists in (b). It also checks the behavioural bound $s(x') \geq 1 - \frac{5}{110 - x'}$. Output: "Tadelis 8.7: (a) no loan, x = 0, (0,0); (b) loan, x = 105, sue every x < 105, (5,5); (c) pure NE = SPE; (d) no loan, x = 0, never sue (tie at x = 0), (0,0); (e) NE with loan and threat to sue, (5,5), and no-loan NE with x <= 100; (f) lender pays up to 5; (g) surplus +10". Tadelis prints no answer.

(f) Common mistakes.

- Reading (c) as "the lender threatens not to sue, so he does not lend". The lender's own off-path plan cannot discipline the lender: he can change his whole strategy, offer and suit rule together. In (b) every Nash equilibrium in pure strategies lends.
- In (d), leaving the tie at $x = 0$ unresolved, or claiming an SPE where the lender sues only at $x = 0$. Then the debtor has no best reply, so that profile cannot be part of an equilibrium.
- In (f) and (g), counting the \$105 fee as a cost of the reform. It is paid only after a default, which never happens in equilibrium.

Answer. (a) SPE (*Not*; $x = 0$), payoffs $(0, 0)$, unique. (b) SPE: offer, repay $x = 105$, sue after every $x < 105$; payoffs $(5, 5)$, unique. (c) None in pure strategies: the only pure Nash equilibrium is the SPE (behavioural NE with off-path suit probability $s(x') \in [1 - \frac{5}{110-x'}, 1)$ exist, with the same outcome). (d) SPE: no loan, $x = 0$, never sue; payoffs $(0, 0)$, unique (the tie at $x = 0$ must go to not suing). (e) Yes: (offer, sue after every default; $x = 105$), payoffs $(5, 5)$, rests on a non-credible threat to sue; so do the other no-loan profiles with planned $x \leq 100$. (f) Yes, up to \$5 (his SPE payoff goes from 0 to 5). (g) Yes: surplus rises from 0 to 10, both gain \$5, and the fee is never paid in equilibrium.

Item 10 (item 40 of the combined list)**: MWG Exercise 9.B.5** Advanced · difficulty 5/5

Source: MWG Exercise 9.B.5, page 302. The statement is reproduced verbatim in the list (item 40); it is not repeated here.

(a) What is asked. For a finite two-player game played simultaneously or with player 1 moving first and observed: (a), (b) count strategies; (c) multiple SPNE imply a payoff tie; (d) without ties for player 2, moving first guarantees player 1 at least any pure Nash payoff of the simultaneous game, in every SPNE (and whether in every Nash equilibrium); (e) examples where (d) fails with ties for player 2 or with a mixed Nash equilibrium.

(b) Method. Strategies as complete plans (MWG §7.D; Lecture 3, slide 25); SPNE and backward induction in finite games of perfect information (MWG §9.B, Prop. 9.B.1 (Zermelo), p. 272, and Prop. 9.B.2; Lecture 3, slides 30 and 35; Lecture 4, slide 17). The sequential game in (b) has perfect information, so its SPNE are the backward-induction profiles; with no ties they are unique. Non-credible threats in Nash equilibrium: Lecture 4, slides 8 and 13.

(c) Solution. Notation. As in the statement, m_i is both the number of actions and (from (c) on) a typical action; write $|M_i|$ for the number to avoid confusion. In the sequential game a strategy of player 2 is a function $s_2 : M_1 \to M_2$.

(a) Simultaneous moves. Each player has one information set, so player i has $|M_i| = m_i$ pure strategies (and the mixed strategies $\Delta(M_i)$, a simplex of dimension $m_i - 1$).

(b) Player 1 first, observed. Player 1 still has one information set: m_1 pure strategies. Player 2 has one information set for each of player 1's moves, each with m_2 actions, so $m_2^{m_1}$ pure strategies (the functions $M_1 \to M_2$; item "Tadelis 7.1").

(c) Multiple SPNE imply a tie. Suppose, to the contrary, that no player is indifferent between any two distinct pairs: $\phi_i(m_1, m_2) \neq \phi_i(m_1', m_2')$ for $i = 1, 2$ whenever $(m_1, m_2) \neq (m_1', m_2')$.

- *Player 2's subgames.* After each m_1, the subgame is a one-player decision; any SPNE must play a best reply there. Since $\phi_2(m_1, \cdot)$ takes m_2 distinct values, the best reply $b(m_1)$ is unique and played with probability 1. So player 2's SPNE strategy is the unique function b.
- *Player 1.* Given b, choosing m_1 yields $\phi_1(m_1, b(m_1))$. The pairs $(m_1, b(m_1))$ differ across m_1 (their first components differ), so by assumption these m_1 payoffs are distinct, and the maximizer m_1^* is unique; any SPNE puts probability 1 on it.

So the SPNE is unique, (m_1^*, b), contradicting multiplicity. Hence if there are multiple SPNE, some player has a tie: (i) or (ii) holds for two distinct pairs. (In fact more is shown: either player 2 has a tie *within* some $\phi_2(m_1, \cdot)$, or player 1 has a tie between two pairs with different m_1.) □

(d) Moving first does not hurt player 1 (in SPNE). Assume (ii) never holds: player 2 has no ties. Let $(\hat{m}_1, \hat{m}_2)$ be a pure Nash equilibrium of the simultaneous game with $\pi_1 = \phi_1(\hat{m}_1, \hat{m}_2)$.

(i) In the simultaneous game $\hat{m}_2$ is a best reply to $\hat{m}_1$: $\phi_2(\hat{m}_1, \hat{m}_2) \geq \phi_2(\hat{m}_1, m_2)$ for all m_2. With no ties, it is the *unique* best reply, so $b(\hat{m}_1) = \hat{m}_2$.

(ii) In any SPNE of the sequential game, player 2 plays $b(m_1)$ after every m_1 (with probability 1, since best replies are unique). In particular she plays $\hat{m}_2$ after $\hat{m}_1$.

(iii) An SPNE is a Nash equilibrium, so player 1's equilibrium payoff is at least what he gets by deviating to $\hat{m}_1$, which is $\phi_1(\hat{m}_1, b(\hat{m}_1)) = \phi_1(\hat{m}_1, \hat{m}_2) = \pi_1$.

So in every SPNE player 1 gets at least π_1 (for every pure Nash equilibrium of (a), hence at least the largest such π_1). □

Not for every Nash equilibrium of (b). A Nash equilibrium of the sequential game may use a non-credible threat by player 2 after moves that player 1 does not make. Example (no ties for player 2; rows player 1, columns player 2, both choose O or F):

	O	F
O	2, 1	0, 0
F	0, −1	1, 2

Player 2's payoffs $1, 0, -1, 2$ are all distinct. The simultaneous game has the pure equilibrium (O, O) with $\pi_1 = 2$ (and (F, F) with 1). In the sequential game the unique SPNE is $(O,\ s_2 = (O \text{ after } O, F \text{ after } F))$, payoff 2 to player 1, as (d) predicts. But $(F,\ s_2 = (F, F))$ is also a Nash equilibrium: against "always F", player 1 gets 0 from O and 1 from F; against F, player 2's on-path choice F is optimal ($2 > -1$). Player 1 gets $1 < 2 = \pi_1$. The threat "F after O" ($0 < 1$ for player 2) is not credible.

(e) Two ways the conclusion of (d) fails.

(e.1) Player 2 has ties. Take the game of the MWG manual, with player 2's payoff depending only on player 1's move:

	L	R
U	0, 1	1, 1
D	0, 0	1, 0

(U, R) is a pure Nash equilibrium of the simultaneous game with $\pi_1 = 1$ (player 1: $1 \geq 1$; player 2: $1 \geq 1$). In the sequential game player 2 is indifferent at both nodes, so every reply is sequentially rational, and $(U, (L \text{ after } U, L \text{ after } D))$ is an SPNE (player 1 gets 0 from either move). Player 1's SPNE payoff is $0 < 1 = \pi_1$.

(e.2) A mixed Nash equilibrium instead of a pure one. Matching pennies:

	L	R
U	1, 0	0, 1
D	0, 1	1, 0

Player 2 has no ties within either row. The simultaneous game has no pure equilibrium; its mixed equilibrium $(\frac{1}{2}, \frac{1}{2})$ gives player 1 $\frac{1}{2}$. In the sequential game player 2 replies R after U and L after D, so player 1 gets 0 whatever he does: the SPNE are $(U, (R, L))$ and $(D, (R, L))$, both with payoff $0 < \frac{1}{2}$. Moving first and being observed is a *disadvantage* when player 1 wants to be unpredictable.

(d) Interpretation. Moving first helps when it is a commitment the follower must respond to (Stackelberg, entry deterrence); (d) makes this precise: if the follower's reply to each move is pinned down, the leader can always reproduce any pure Nash outcome of the simultaneous game by playing his part of it, so he cannot lose by moving first. The two failures in (e) mark the limits: if the follower is indifferent, her tie-breaking can punish the leader at no cost to herself; and if the leader's simultaneous-move advantage comes from being unpredictable (mixing), observation destroys it. The Nash-but-not-SPNE example in (d) is the Lecture 4 point once more: threats off the path can make the leader look worse off, but they are not credible.

(e) Verification. `check_B_L04.py` enumerates all pure SPNE (a strategy $s_2 : M_1 \to M_2$ that best replies in every subgame, and a best m_1) of 3000 random games with $m_1, m_2 \in \{2, 3\}$ and integer payoffs in $\{0, \ldots, 4\}$ (fixed seed). In each of the 1795 games with several SPNE a tie (i) or (ii) is present; in each of the 166 games without ties for player 2, every SPNE gives player 1 at least the payoff of every pure Nash equilibrium of the simultaneous game. It confirms the counterexample in (d) (unique SPNE pays 2, the Nash equilibrium $(F, (F, F))$ pays 1), (e.1) (SPNE $(U, (L, L))$ pays 0 against $\pi_1 = 1$) and (e.2) (SPNE $(U, (R, L))$, $(D, (R, L))$ pay 0; mixed value $\frac{1}{2}$). Output line: `mwg-9-b-5: (c) on 1795 random games with multiple SPNE, (d) on 166 games; examples (d), (e1), (e2) confirmed`. *Comparison with the MWG solutions manual* (Exercise 9.B.5, pp. 9-2 to 9-4; label OCR'd as "9.8.5"): (a) m_i pure strategies; (b) m_1 and $m_2^{m_1}$; (c) proof by contradiction, as here; (d) the same argument, and "low payoffs for player 1 could be sustained in a NE with incredible threats by player 2"; (e) the same two examples (the indifferent follower of Fig. 9.B.5(e.1) and matching pennies, Fig. 9.B.5(e.2)). **Agreement on every part.** One slip in the manual (or its OCR): the SPNE of the pennies game are written "$(U, (R, L))$ and $(L, (R, L))$"; the second must be $(D, (R, L))$, since L is not a move of player 1.

(f) Common mistakes.

- Giving player 2 $m_1 m_2$ strategies in (b) (pairs) instead of $m_2^{m_1}$ (functions).
- In (d), forgetting that the argument needs player 2's best reply to $\hat{m}_1$ to be *unique*: with a tie she might answer $\hat{m}_1$ differently in the sequential game, which is exactly example (e.1).
- Claiming that moving first is always an advantage: (e.2) shows it can be strictly worse.

Answer. (a) m_1 and m_2. (b) m_1 for player 1, $m_2^{m_1}$ for player 2. (c) Without ties player 2's reply to each move is unique and then player 1's best move is unique, so the SPNE is unique; multiple SPNE therefore require (i) or (ii). (d) In any SPNE player 2 answers $\hat{m}_1$ with her unique best reply $\hat{m}_2$, so player 1 can secure π_1 by playing $\hat{m}_1$; not for every Nash equilibrium (e.g. $(F, (F, F))$ in the example above gives $1 < 2$). (e) Indifferent follower: $(U, (L, L))$ is an SPNE with payoff $0 < 1$; matching pennies: SPNE payoff $0 <$ mixed Nash payoff $\frac{1}{2}$.